FIN 3080 Tutorial 3

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Agenda

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  - ✓ Assignment 2 Overview
  - ✓ Data Preparation: CSMAR & Provided Data
  - ✓ Core Task Breakdown
  - ✓ Report Structure & Data Visualization
  - ✓ Tips

Assignment 2 Overview

- This assignment asks you to compare three bubbles:
  - ✓ 2015 episode: 2014-06-19 to 2015-06-12
  - ✓ 2021 episode: 2019-01-04 to 2021-02-18
  - ✓ current episode: 2024-09-18 to 2026-03-09
- You need to:
  - ✓ Compute all required indicators for the 2015 and 2021 episodes
  - ✓ Compute the same indicators for the current episode
  - ✓ Compare indicators across episodes
  - ✓ Draw a conclusion on whether the current episode is in the early / mid / late stage of a bubble
  - ✓ Plot time-series trend figures for the current episode

Assignment 2 Overview

- What is the goal of this assignment?
  - ✓ The purpose is not just calculation. The deeper goal is to understand:
    - what a market bubble looks like in data,
    - how bubble-like episodes differ from normal markets,
    - and whether the current market has reached historical extremes.
- So your report should not stop at:
  - ✓ “Indicator A = x.”
  - ✓ “Indicator B = y.”
- You should also discuss like:
  - ✓ Does this indicator usually rise or fall in bull markets?
  - ✓ Is the current episode already comparable to 2015 / 2021?
  - ✓ Which indicators suggest overheating? Which stage is the market in?

A Useful template for each indicator

- For almost every indicator, your report can follow the same logic:
    1. In the 2015 episode, the indicator equals …
    2. In the 2021 episode, the indicator equals …
    3. In the current episode, as of 2026-03-09, the indicator equals …
    4. Comparing the three episodes, we see that …
    5. This suggests that the current market is …

Suggested Workflow of This Assignment

Suggested Workflow

1. Download / Organize all required datasets
2. Define the three bubble intervals clearly
3. For each interval and each indicators, compute
    a) start value
    b) end value
    c) peak value if required
    d) corresponding ratios / changes
4. For the current episode, compute running indicators over time: from 2024-09-18 to each date up to 2026-03-09
5. Compare across episodes and write interpretation
6. Conclude the stage of the current episode

1. Download / Organize all required datasets

- You need to download from CSMAR:
  - ✓ Daily total A-share trading value (Main and GEM board): *China Stock Market Series – Stock Trading* database
  - ✓ SSE Composite Index daily price: *China Stock Market Series – Market Index* database
  - ✓ Daily margin trading and securities lending balance: *China Stock Market Series – Margin Trading and Securities Lending* database
  - ✓ Daily 10-year government bond yield: *China Bond Market Series – Bond Market Trading – Interest Rate and Yield* database.

1. Download / Organize all required datasets

- Before calculation, you need to unify your data structure. (refer to 2nd tutorial)
  - ✓ You are using multiple datasets with:
    - Different frequencies
    - Different data formats
    - Different units
  - ✓ So before doing any indicator calculation, make sure you:
    - Covert dates into proper date format
    - Sort data by date
    - Check for duplicates / missing values
    - Verify measurement units

2. Overview of the seven required indicators

1. Trading value ratio
2. Maximum trading value ratio within the bull market
   - And compare SSE index return before vs. after trading value peak
3. Margin trading balance ratio
4. New monthly issuance of equity-oriented funds ratio
5. CSI300 ERP: Level change and Percentile change
6. CSI300 Dividend Yield – 10Y Gov Bond Yield: Level change and Percentile change
7. Price change of representative stocks: East Money, Hundsun, and best-performing brokerage stock

2. Overview of the seven required indicators

Indicator 1: Trading Value Ratio

1. Definition

$$Trading\ Value\ Ratio = \frac{\mathrm{A} - share\ total\ trading\ value\ at\ the\ end}{A - share\ total\ trading\ value\ at\ the\ start}$$

2. How to calculate

- For each episode:
  a) Sum the trading value in different markets together
  b) Find the trading values on the start and end date, compute the ratio

2. Overview of the seven required indicators

Indicator 1: Trading Value Ratio

3. For the current episode, you also need a time-series trend chart:
   - X-axis: date
   - Y-axis: trading value relative to the value on 2024-09-18
   - So for each date t,

$$\frac{Trading\ value\ at\ t}{Trading\ value\ at\ 2024-09-18}$$

Remark: Is your trading value for the entire A-share market?

2. Overview of the seven required indicators

Indicator 2: Maximum Trading Value Ratio

1. Definition

$$\mathit{Max\ Trading\ Value\ Ratio}$$
$$= \frac{\max(\mathrm{A} - \mathit{share\ total\ trading\ value\ during\ the\ episode})}{A - \mathit{share\ total\ trading\ value\ at\ the\ start}}$$

2. For the current episode, this assignment says suppose the maximum trading value of the current episode has already occurred. So you can simply use the highest observed trading value between 2024-09-18 and 2026-03-09.

2. Overview of the seven required indicators

Indicator 2: Maximum Trading Value Ratio

3. After finding the date of trading value peak, compare the SSE Composite Index return:
   - Before trading value peak

$$\frac{Price_{peak}}{Price_{start}} - 1$$

   - After trading value peak

$$\frac{Price_{end}}{Price_{peak}} - 1$$

4. Question to discuss: does most of the price appreciation happen before or after trading activity peaks?

2. Overview of the seven required indicators

Indicator 2: Maximum Trading Value Ratio

5. Workflow:
   - In the trading value data, identify the maximum trading value within the interval, and calculate the ratios. Also record the peak date.
   - In the SSE Composite dataset, obtain start-date, peak-date, end-date index levels. Compute the returns and compare.
6. Running peak series for the current episode:

$$\frac{\max(Trading\ value\ from\ start\ to\ date\ t)}{Trading\ value\ at\ start}$$

So for each date t:

- look back from 2024-09-18 to t and take the historical maximum so far.
- Divide by the initial trading value. This gives a non-decreasing line.

2. Overview of the seven required indicators

Indicator 3: Margin Trading Balance Ratio

1. Definition

$$\begin{aligned} &Margin\ Balance\ Ratio \\ &= \frac{Margin\ trading\ \&\ securities\ lending\ balance\ at\ the\ end}{Margin\ trading\ \&\ securities\ lending\ balance\ at\ the\ start} \end{aligned}$$

2. How to calculate: very similar to the first indicator
3. Remark: also take care of summing all values in different markets together!

2. Overview of the seven required indicators

Indicator 4: New Issuance of Equity-Oriented Funds Monthly

1. Definition

$$Fund\ Issuance\ Ratio$$

$$= \frac{Newly\ established\ equity - oriented\ fund\ shares\ in\ end\ month}{Newly\ established\ equity - oriented\ fund\ shares\ in\ start\ month}$$

2. Because the current episode has not ended yet, **use February 2026 as the end month**. If there is **no value in the start month or end month**, use the **nearest** available month **within** the interval.
3. The dataset provided is daily, but the assignment asks for monthly issuance comparison.

2. Overview of the seven required indicators

Indicator 4: New Issuance of Equity-Oriented Funds Monthly

4. So your task is essentially:
    - Sum the issuance value at each date within one month together.
    - Identify the relevant start month and end month.
    - Obtain the issuance value for those months.
    - Compute the ratio.
    - Build the monthly time-series figure.

2. Overview of the seven required indicators

Indicator 5: CSI300 ERP

1. Definition

$$ERP = \frac{1}{PE_{CSI300}} - BondYield_{10Y}$$

2. Remark: check the unit of 10-year government bond yield!
3. You must analyze two things:
   - Level change: compare ERP at start and end
   - Percentile change: evaluate where the start ERP and end ERP sit in a historical distribution

2. Overview of the seven required indicators

Indicator 5: CSI300 ERP

4. The assignment asks you to use an empirical CDF based on historical data before the bubble period.
   - For the 2015 episode, use data from 2005 to 2013-12-31
   - For the 2021 episode, use data from 2005 to 2018-12-31
   - For the current episode, use data from 2005 to 2023-12-31

   Then:
   - Construct the historical ERP series
   - Compute the empirical distribution
   - Evaluate the percentile of ERP at the start and end of the episode

2. Overview of the seven required indicators

Indicator 6: CSI300 Dividend Yield – 10Y Bond Yield

1. Definition

$$Dividend\ Spread = CSI300\ Dividend\ Yield - BondYield_{10Y}$$

2. Again, you need to analyze level change and percentile change. Use the same idea as ERP.

2. Overview of the seven required indicators

Indicator 7: Price Change of Representative Stocks

1. You need to compute price changes for three types of stocks:
    - East Money
    - Hundsun Technologies
    - A brokerage stock: choose the **single** brokerage stock with the largest price increase in the interval
2. For each stock, price change over the bubble interval is:

$$\frac{Price_{end}}{Price_{start}} - 1$$

2. Overview of the seven required indicators

Indicator 7: Price Change of Representative Stocks

3. For the brokerage stock:
   - Search all brokerage stocks in the provided representative stock dataset. And compute interval return for each.
   - Choose the one with the largest return. And report the stock code and the return.
4. For the current episode time-series figure, you don't need to find the best broker stock repeatedly in rolling windows. Instead, first find the brokerage stock with the beast performance over the whole current interval. Then plot the price change series using $\frac{Price_t}{Price_{start}} - 1$ for dates from 2024-09-18 to 2026-03-09.

3. How to think about “bubble stage” ?

You should synthesize all indicators. Possible logic:

1. Early stage:
   - Trading activity not yet expanded much.
   - Leverage mild.
   - Fund issuance muted.
   - Valuation still attractive.
2. Mid stage:
   - Trading activity clearly stronger.
   - Some sentiment and leverage buildup.
   - Valuations less attractive but not extreme.

3. How to think about “bubble stage” ?

You should synthesize all indicators. Possible logic:

3. Late Stage:

   - Explosive turnover.
   - Heavy leverage.
   - Rapid fund issuance.
   - Valuation indicators compressed to historical extremes.
   - Representative stocks surge abnormally.

4. Possible Stata commands

- use, merge, append
- gen, replace, format
- collapse
- egen
- bysort
- tsset or date sorting
- Graph commands: twoway line, graph bar

Thank you!